

PARTS LIST

Semiconductors:

- IC1 - 7805 5 voltage regulator
- IC2 - ATmega32 MCU
- LED1-LED5 - 5mm LED
- LCD1 - 16x2 LCD
- T1 - BC547 npn transistor

Resistors (all 1/4-watt, ±5% carbon):

- R1, R4-R7 - 330-ohm
- R2 - 10-kilo-ohm
- R3 - 470-ohm
- R8-R11 - 1-kilo-ohm
- VR1 - 10-kilo-ohm trimpot

Capacitors:

- C1, C4 - 0.01µF ceramic disk
- C2, C3 - 20µF ceramic disk
- C5 - 0.1µF ceramic disk

Miscellaneous:

- XTAL1 - 16MHz crystal oscillator
- PZ1 - Piezo buzzer
- S1-S5 - Tactile switch
- JACK - DC jack connector
- CON1, CON6 - 1-pin connector
- CON2, CON4, CON7, CON9 - 8-pin connector
- CON3, CON5 - 6-pin connector
- CON8, CON10-CON12 - 4-pin connector
- 12V, 500mA power adaptor (AC-DC)

```

NOT_ON_TIMER, /*1*/
NOT_ON_TIMER, /*2*/
TIMER0, /*3 8 bit*/
NOT_ON_TIMER, /*4*/
NOT_ON_TIMER, /*5*/
NOT_ON_TIMER, /*6*/
NOT_ON_TIMER, /*7*/
NOT_ON_TIMER, /*8 - port D */
NOT_ON_TIMER, /*9*/
NOT_ON_TIMER, /*10*/
NOT_ON_TIMER, /*11*/
TIMER1B, /*12*/
TIMER1A, /*13*/
NOT_ON_TIMER, /*14*/
TIMER2, /*15 8 bit*/
NOT_ON_TIMER, /*16 - port C */
NOT_ON_TIMER, /*17*/
NOT_ON_TIMER, /*18*/
NOT_ON_TIMER, /*19*/
NOT_ON_TIMER, /*20*/
NOT_ON_TIMER, /*21*/
NOT_ON_TIMER, /*22*/
NOT_ON_TIMER, /*23*/
NOT_ON_TIMER, /*24 - port A */
NOT_ON_TIMER, /*25*/
NOT_ON_TIMER, /*26*/
NOT_ON_TIMER, /*27*/
NOT_ON_TIMER, /*28*/
NOT_ON_TIMER, /*29*/

```

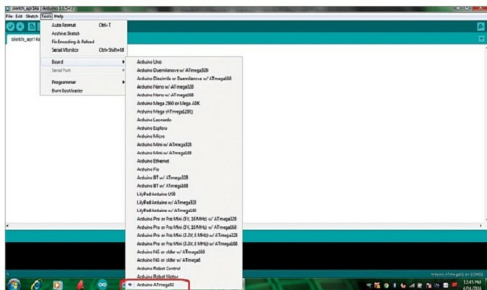


Fig. 5: Arduino ATmega32 board under Tools menu

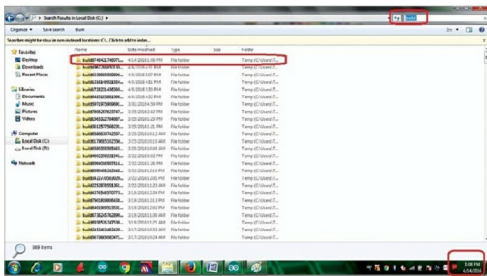


Fig. 6: Hex file for Arduino ATMEGA32 under build in C://

```

NOT_ON_TIMER, /*30*/
NOT_ON_TIMER, /*31*/
};

Search for hardware\arduino\
cores\arduino directory. Open
Arduino.h file and add the following
line after the last timer-defined line,
that is, after #define TIMER5C 17,
which comes after line number 184.

#define TIMER0 18

```

Open Tone.cpp file in the cores directory and add/modify the following to suit Timer2 of ATmega32.

After line number 39, add defined(__AVR_ATmega32__) as shown below:

```

#if defined(__AVR_ATmega8__) || defined(
AVR_ATmega128__) || defined(__AVR
ATmega32__)

```

After line number 56, add !defined(__AVR_ATmega32__) as shown below:

```

#if defined(__AVR_ATmega8__) ||
!defined(__AVR_ATmega32__)
After line number 96, add
defined(__AVR_ATmega32__) as
shown below:
#elif defined(__AVR_ATmega8__) ||
defined(__AVR_ATmega32__)

```

Open WInterrupts.c in the same cores directory and add following for ATmega32 int2 after line number 147:

```

#elif defined(MCUCSR) && defined(ISC2)
MCUCSR = (MCUCSR & ~(1
<< ISC2)) | ((mode & 1) << ISC2);
GICR |= (1 << INT2);

```

After line number 324 add: